



MASTER THESIS

Match the Player, Not the Game: An Exploratory Study of Personality-Adaptive Tutorial Design in Souls-like Games.

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DECLARATION OF AUTHORSHIP

I, Louis RIVES-LEHTINEN, declare that this thesis titled, “Match the Player, Not the Game: An Exploratory Study of Personality-Adaptive Tutorial Design in Souls-like Games” and the work presented are my own. I confirm that:

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- Where I have consulted the published work of others, this is always clearly attributed.
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“Working through a study whose hypothesis was not supported has taught me more about research than a confirmation would have. I leave this project with a clearer sense of what data can and cannot say, and real respect for the work of asking the right question of it.”

Louis RIVES-LEHTINEN

ABSTRACT

Academy for AI, Games and Media

Master of Game Technology

Match the Player, Not the Game: An Exploratory Study of Personality-Adaptive Tutorial Design
in Souls-like Games.

By Louis RIVES-LEHTINEN

Souls-like games face high early abandonment despite critical success, and existing onboarding remains one-size-fits-all even though identical tutorials produce widely different outcomes across players. Personality psychology has long predicted that matching instructional style to learner traits improves outcomes in education, workplace training, and clinical settings, yet whether the same principle applies to game tutorials remains untested. This study tested whether matching tutorial intensity (minimal versus explicit) to player Openness improves the initial Souls-like experience among Western players. A quasi-experimental 2×2 between-subjects design (High/Low Openness × Minimal/Explicit Tutorial) was run with 29 participants playing *Lies of P* up to its first boss. Openness was measured with the BFI-10, a brief Big Five personality questionnaire; primary outcomes were self-reported enjoyment, perceived competence, and intention to continue. The predicted personality-by-tutorial interaction did not emerge, and Openness as a stand-alone moderator failed to produce significant effects on any primary outcome. The one significant tutorial effect went the opposite way to what was expected: players given only the basic in-game tutorial reported feeling more capable than players given a detailed walkthrough. By chance, the random assignment also placed far more experienced Souls-like players in the less-guided group, so part of that difference may reflect prior experience rather than the tutorial itself. Openness alone is too narrow to drive tutorial personalisation; a wider personality model that includes Conscientiousness, stratified randomisation on gaming experience, and an in-game manipulation are needed before the personality-matching hypothesis can be properly evaluated.

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LIST OF ABBREVIATIONS

Low-O	Low Openness
High-O	High Openness
Low-C	Low Conscientiousness
High-C	High Conscientiousness
UX	User Experience
BFI-10	Big Five Inventory 10
BH	Benjamini-Hochberg

To my family, and to my grandparents most of all — who have been there for me through everything, including the parts that have asked the most of them.

1 INTRODUCTION

1.1 THE CHALLENGE OF GETTING PLAYERS THROUGH THE FIRST HOUR

Imagine two friends, Cody and May, both buying *Dark Souls* for the first time. Within the first hour, Cody drops their controller in frustration and never plays again. May dies repeatedly but keeps coming back, eventually beating the first boss after 20 attempts. What's the difference? The game is identical. The difficulty is the same. Yet their experiences and their decisions to continue or quit are fundamentally different.

This scenario illustrates the central problem this research addresses: early player abandonment in challenging games. Understanding why some players persist while others quit could transform how games introduce players to their mechanics, especially in notoriously difficult genres.

1.2 THE PROBLEM: EARLY ABANDONMENT IN SOULS-LIKE GAMES

Souls-like games represent a subgenre of action role-playing games pioneered by Japanese developer FromSoftware with *Demon's Souls* (2009) and the *Dark Souls* series (2011-2016). These games are characterized by high difficulty with death as the primary learning mechanism—players die frequently (9 billion deaths recorded in *Elden Ring*'s first year; Bandai Namco Entertainment, 2023)—stamina-based combat requiring strategic engagement, sparse checkpoints where death costs significant progress, and environmental storytelling that emphasizes player interpretation over explicit narrative. Cryptic, fragmented lore presents NPCs who speak in riddles and objectives that remain deliberately unclear, with story emerging through environmental clues rather than cutscenes (Juul, 2013; Vella, 2015).

Despite their critical acclaim and dedicated fanbase, Souls-like games face a significant challenge: high early abandonment rates. While exact figures vary by title, industry data suggests that a substantial proportion of players never progress beyond the first major area or boss encounter. This represents both a commercial challenge (lost potential audience) and a design puzzle (how to introduce such demanding gameplay without overwhelming newcomers).

Current onboarding approaches remain largely one-size-fits-all. Every player receives the same tutorial, the same introduction, the same initial guidance—regardless of their individual

characteristics, preferences, or learning styles. Yet as the Cody and May scenario demonstrates, identical onboarding produces wildly different outcomes. Some players thrive on minimal guidance and discovery-driven learning, while others become frustrated and abandon the game entirely when confronted with ambiguity and uncertainty.

1.3 WHY PERSONALITY MIGHT MATTER: A NEW APPROACH TO ONBOARDING

Tutorial design represents a fundamental user experience (UX) challenge in game development. While UX research traditionally focuses on usability, ensuring interfaces are learnable and error-free (Nielsen, 1994), game onboarding introduces complexity. Games intentionally create challenge and uncertainty as core mechanics, so "good UX" cannot simply minimize difficulty. Instead, player experience (PX) research examines balancing challenge with accessibility (Andersen et al., 2012). Unlike traditional UX problems with clear solutions (e.g., clearer buttons, better navigation), personality-adaptive onboarding requires matching instructional design to psychological traits—presenting identical mechanics through different frameworks based on predicted player needs.

Traditional game design assumes uniform player responses to difficulty. However, personality psychology suggests that stable individual differences in curiosity, ambiguity tolerance, and persistence fundamentally alter how players experience identical games (McCrae & Costa, 1987). This research proposes personality acts as a perceptual filter—transforming identical challenges into different subjective experiences.

This research focuses on Openness to Experience from the Big Five personality model—measuring creativity, curiosity, and comfort with ambiguity (detailed in Section 2.3). The connection to Souls-like games is direct: these games require characteristics defining high Openness. High Openness players should experience Souls-like design as intriguing mystery; low Openness players should experience it as frustrating uncertainty.

This suggests a solution: personality-adaptive onboarding. Games could assess player personality and match tutorial design accordingly. High Openness players receive minimal guidance preserving discovery; low Openness players receive explicit instruction reducing ambiguity. Both face identical difficulty—same enemies, same mechanics, same challenge—but with different instructional support during the critical initial period.

Edwards' (2008) Person-Environment Fit theory provides the mechanism: optimal outcomes occur when environmental demands match personal resources. Matched conditions provide

appropriate support without over-scaffolding or under-supporting. Ryan and Deci's (2000) Self-Determination Theory suggests High Openness players prioritize autonomy over immediate competence, while Low Openness players prioritize competence over autonomy, creating a paradox where explicit tutorials intended to help may harm High Openness players by violating autonomy needs.

If successful, this approach could expand accessibility without compromising core difficulty. Accessibility improvements wouldn't require difficulty reduction, which fans resist as compromising genre identity. Instead, personalized scaffolding maintains challenge while reducing uncertainty for players needing clarity and preserving discovery for players valuing autonomy.

1.4 THE RESEARCH QUESTION

Despite extensive research on game difficulty, player motivation, and tutorial design, no studies examine personality-tutorial interactions. This thesis examines: **Does matching tutorial type (minimal vs. explicit) to player Openness level improve subjective experience and behavioural persistence during initial Souls-like gameplay among Western players?**

This pilot study (N=24-32) will employ an existing Souls-like game with modified tutorial conditions and focused dependent measures to ensure practical feasibility for a master thesis project. The research tests whether individual differences in personality traits interact with tutorial design to predict player experience and persistence during a single gameplay session. While sample size and single-session design limit generalizability, this investigation can establish proof-of-concept for personality-adaptive onboarding, demonstrating whether the approach warrants larger-scale investigation.

Four hypotheses guide the research: (1) High Openness players will show greater persistence overall, reflecting natural compatibility with Souls-like genre characteristics; (2) Explicit tutorials will produce more positive subjective experience overall by reducing initial frustration; (3) Matched players (High-O with minimal tutorial, Low-O with explicit tutorial) will show better outcomes than mismatched players—the critical interaction effect; (4) Perceived difficulty will mediate the relationship between personality-tutorial match and behavioural persistence.

The following chapters develop the theoretical foundation for these hypotheses (Chapter 2), detail the experimental methodology (Chapter 3), present findings (Chapter 4), interpret results

within theoretical frameworks (Chapter 5), and synthesize contributions with implications for theory and practice (Chapter 6).

1.5 THESIS STRUCTURE

This is the practical problem we are addressing: high-difficulty games like those in the Souls-like genre experience substantial early player abandonment, yet we lack evidence-based approaches for matching onboarding strategies to individual player needs. While game designers recognize that different players respond differently to challenge, current onboarding approaches remain largely one-size-fits-all. This research investigates whether personality assessment before gameplay can predict which tutorial approach, minimal discovery-based or explicit instruction-based, will optimize each player's initial experience and increase the likelihood of continued engagement. If successful, this approach could transform how challenging games introduce their mechanics, expanding accessibility without compromising the core difficulty that defines the genre.

2 LITERATURE REVIEW

2.1 INTRODUCTION: BUILDING THE THEORETICAL FRAMEWORK

Answering why some players stick with a hard game while others give up requires bringing together research areas that do not usually talk to each other. Personality psychology explains stable differences between people. Game studies documents what makes Souls-like design distinctive. Learning theory shows how different kinds of instruction shape early engagement. Person-Environment Fit theory connects all three by explaining what happens when support does or does not match what someone needs. None of these fields has yet looked at how they interact in a gaming context. This review makes the case that they should.

2.2 FROM GAMING TYPOLOGIES TO TRAIT PSYCHOLOGY

2.2.1 Limitations of Gaming-Specific Typologies

Attempts to categorize players are nothing new. Bartle's (1996) four player types — Achievers, Explorers, Socializers, Killers — gave designers a useful shorthand for thinking about player differences in multiplayer games. Yee's (2006) motivation components (Achievement, Social, Immersion) took this further in MMORPGs with more empirical backing.

The problem is that these typologies fall apart when tested properly. When Yee surveyed 3,200 MMORPG players, the data did not behave as Bartle's model predicted: motivations correlated with each other rather than sitting in opposition ($r = .10$ to $.41$), and the underlying structure looked nothing like the four-type framework. More importantly, gaming-specific typologies exist in a bubble: they have not been tested outside gaming, they lack the rigorous development of established personality frameworks, and they cannot connect to the broader body of research on how people differ (Yee, 2006). They can describe what players do. They cannot explain why.

2.2.2 Big Five: Strengths and Gaming-Specific Critiques

The Big Five (OCEAN) model is a different matter. It is the most thoroughly tested personality framework in psychology. Its five dimensions — Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism — are around 50% genetic, stay stable across a person's life, and reliably predict meaningful outcomes (Bouchard & Loehlin, 2001; John et al., 2008; McCrae & Costa, 1987).

Table 1*Comparison of Personality Frameworks for Gaming Research*

Framework	Type	Key Features	Gaming Application	Limitations
<i>Big Five/OCEAN</i>	General personality	5 traits, extensively validated, cross-cultural	Broad personality-game correlations	Small effect sizes with gaming behavior (Akbari et al., 2021)
<i>HEXACO</i>	General personality	Adds Honesty-Humility as 6th factor	Differentiates gaming populations (Abbasi et al., 2020)	Still general, not gaming-specific
<i>BrainHex</i>	Gaming-specific	7 neurobiological archetypes	Direct gaming motivation mapping	Only 2/7 types reliably validated (Busch et al., 2016)
<i>Hexad</i>	Gamification-specific	6 types based on SDT	Gamification contexts (Tondello et al., 2016)	Limited to gamification, not core gaming
<i>Gaming Typologies (Bartle, Yee)</i>	Gaming-specific	Player types/motivations	MMORPG design	Lack psychometric validation, assume discrete types

The trade-off is honest. Gaming-specific models might predict gaming behaviour more precisely, but most of them are not well-validated — BrainHex, for example, has only two of its seven types holding up under scrutiny (Busch et al., 2016). And as Kirchner-Krath et al. (2024) note, gaming typologies tend to collapse into roughly five dimensions when tested empirically anyway. For a pilot study, the case for Big Five is pragmatic: there are validated short measures (BFI-10), it allows comparisons across research areas, and the link between Openness and exploratory behaviour is well established (DeYoung, 2013). Its relatively modest associations with gaming behaviour (Akbari et al., 2021) are worth acknowledging, but they are not a reason to abandon it.

2.3 OPENNESS TO EXPERIENCE: THE DISCOVERY-MOTIVATION SUBSTRATE

2.3.1 Conceptual Foundation

Openness to Experience measures creativity, curiosity, preference for novelty, comfort with ambiguity, and intellectual engagement (John et al., 2008). High Openness individuals are creative, imaginative, enjoy abstract ideas, seek novel experiences, and tolerate uncertainty (McCrae & Sutin, 2009). They prefer exploration over routine, value aesthetic experiences, and engage comfortably with ambiguous situations. Conversely, Low Openness individuals are practical, traditional, prefer routine and clear structure, feel uncomfortable with ambiguity, and seek predictability. They value concrete over abstract, favor familiar over novel, and experience anxiety in uncertain situations.

2.3.2 Why Openness Matters for Souls-Like Games

Once you understand what Openness actually measures, the connection to Souls-like games is hard to miss. These games ask players to deal with vague objectives, cryptic storytelling, learning through repeated failure, and finding meaning in environmental details rather than explicit instruction. Every one of these demands maps directly onto the difference between high and low Openness. The ambiguity that feels like an intriguing mystery to a high Openness player feels like pointless frustration to a low Openness player. Trial-and-error learning that feeds one player's curiosity feeds another's anxiety. Environmental storytelling that rewards reading between the lines, which Vella (2015) identifies as central to the genre's appeal, is exactly the kind of task high Openness people are drawn to.

The research backs this up. Wartberg et al. (2023) found that lower Openness is associated with more problematic gaming patterns, while Bonnaire and Baptista (2019) showed that Openness predicts a preference for complex, unconventional game content — which describes Souls-like games well. The pattern is not limited to gaming either: Manolika (2023) found that Openness predicts preferences for complex, unconventional content in books and film too, suggesting this is a general disposition rather than something specific to one genre.

2.4 CONSCIENTIOUSNESS AS FUTURE RESEARCH DIRECTION

Conscientiousness — how self-disciplined, persistent, and willing to practise someone is (Roberts et al., 2009) — is clearly relevant to Souls-like success. Getting through these games requires grinding through failure over and over. A player who stops after ten deaths is not necessarily unskilled; they may simply be less disposed toward that kind of repetitive effort.

Openness and Conscientiousness likely interact to create distinct player types, and the complete picture probably requires both. But testing interactions between two traits needs a larger sample than $N=20-30$. This pilot is focused on Openness; Conscientiousness is the natural next step for a larger follow-up study.

2.5 CULTURAL CONTEXT: WESTERN SAMPLE JUSTIFICATION

This research focuses exclusively on Western markets (North America and Europe) for a clear methodological reason. Cultural background affects both personality and how people prefer to learn, and a small pilot study cannot untangle these influences at the same time. Western cultures tend to favour self-directed, low-guidance learning, while East Asian cultures tend toward more structured, guided approaches (Chen, 2014; Swierczek & Bechter, 2010). McCrae et al. (2005) show that cultural background even predicts where people score on personality dimensions, which means a mixed-culture sample would introduce a second explanatory variable into a study that already has limited statistical power. Limiting the sample to Western participants is not an oversight; it is a necessary step toward getting a clean result. Whether the pattern holds in Eastern markets is a question for future research.

2.6 EVIDENCE FROM OTHER CONTEXTS

Souls-like games push challenge-based design to an extreme, but other genres face similar problems. Andersen et al.'s (2012) study of 45,000 players across three games of varying complexity found that tutorials improved player retention by 29% in complex games, while having no meaningful effect in simpler ones. In other words, the harder the game, the more it matters how you introduce it. What the broader literature has not produced, however, is any theory that works across genres — research on roguelikes, fighting games, and simulation games has largely stayed genre-specific, with little insight into why tutorial design affects retention in the first place (see Chapter 5 for comparison).

Beyond gaming, tailoring instruction to personality is not a new idea, it simply has not been applied to games yet. In education, high Openness students consistently do better with discovery-based learning, while low Openness students benefit more from direct, structured instruction (Komarraju et al., 2011). The same pattern appears in workplace training: high Openness employees engage better with open-ended, self-directed development, while low Openness employees get better results from step-by-step programmes (Major et al., 2006). It

even shows up in healthcare — doctors who adjust how they explain options based on patient Openness achieve significantly better treatment compliance (Cousin et al., 2013).

Taken together, these findings point to something more general: people learn and engage better when the way information is structured matches how they are naturally inclined to process it. This principle holds across education, professional training, and clinical settings. There is no obvious reason why game tutorials would be exempt from it.

2.7 ONBOARDING STRATEGIES: HOW GAMES INTRODUCE CHALLENGE

2.7.1 Tutorial Intensity: The Primary Experimental Manipulation

Tutorial intensity — how explicitly a game guides players through its mechanics — is the main variable in this study. It is chosen because it sits at the centre of the personality-match hypothesis and is practically manageable to manipulate. The two conditions run from minimal (the typical Souls-like approach: environmental cues, no explicit instruction) to explicit (step-by-step guidance with tooltips). An existing Souls-like game with modified tutorial conditions makes this feasible without having to build something from scratch.

2.7.2 The Personality-Tutorial Matching Hypothesis and Its Mechanism

The core prediction is that tutorial effectiveness depends on who is receiving it. For high Openness players, being given explicit instruction is not a neutral experience — it actively undermines the game by removing the ambiguity and freedom they are looking for. For low Openness players, minimal guidance is equally not neutral — it creates uncertainty they cannot sit comfortably with, making it hard to tell whether they are failing because of the game's design or their own misunderstanding. Clear instruction removes that particular source of anxiety, and gives them the foundation they need to actually develop competence.

Edwards' (2008) Person-Environment Fit theory formalises this. When the support an environment provides matches what someone actually needs, outcomes improve. When support exceeds what someone needs, it costs them their sense of agency. When support falls short of what they need, they cannot develop competence. For tutorials: a high Openness player matched with minimal guidance gets what they need; a high Openness player given explicit instruction gets more than they need, and that excess is a problem. The reverse is true for low Openness players.

Ryan and Deci's (2000) Self-Determination Theory explains the psychological mechanism behind this. High Openness players value autonomy — the freedom to explore — more than they value feeling immediately competent. They will trade clarity for freedom. Low Openness players value competence — knowing what they are doing — more than they value freedom. They will trade freedom for clarity. This creates the central paradox: a tutorial designed to help may actively harm high Openness players, not because it fails at its job, but because it does that job too thoroughly.

2.8 DIFFICULTY AND PERSONALITY: WHY IDENTICAL GAMES FEEL DIFFERENT

2.8.1 Objective vs. Perceived Difficulty

There is an important distinction between objective and perceived difficulty. Objective difficulty is fixed: enemy health, damage values, move patterns — the same numbers for everyone who plays. Perceived difficulty is not fixed. It is shaped by how skilled you are, how confident you feel, how you explain your failures to yourself, and the frame your personality puts around the experience. The same enemy can feel like a fair challenge to one player and an unfair wall to another — not because the encounter changed, but because they did not encounter it the same way.

This is what the Cody and May scenario is really about. Their objective difficulty is identical. Their experience of that difficulty is not, because personality changes how the same input lands.

2.8.2 Static Difficulty Philosophy

Unlike many modern games employing dynamic difficulty adjustment, Souls-like games use static difficulty—it never adjusts based on performance. Director Hidetaka Miyazaki (2009) explains: "Having the game be 'difficult' was never the goal. What we set out to do was strictly to provide a sense of accomplishment." Dynamic difficulty adjustment would undermine that: if the game secretly gets easier when you struggle, you can never be certain that you actually got better.

This creates a specific design constraint. Since the difficulty cannot adapt to the player, something else must. Tutorial design is the logical lever: the challenge stays fixed, but the guidance around it can change.

2.8.3 The Evolving Souls-Like: From Pure Challenge to Nuanced Accessibility

Miyazaki's commitment to fixed difficulty has not stopped the genre from evolving. *Elden Ring* (2022) introduced several features that let players effectively manage their own experience without the game acknowledging it: Spirit Summons provide AI assistance, the open-world structure allows players to overlevel before hard encounters, and multiplayer passwords let friends help regardless of level gaps. Other developers have taken more direct approaches — explicit difficulty options, expanded co-op features. What this reveals is a distinction the genre has always contained: core mechanical difficulty (enemy damage, move complexity) stays static, while structural accessibility (alternative routes, optional help) can vary without breaking the fundamental challenge. Vella (2015) argues this preserves the genre's "ludic sublime" — the feeling of confronting something that seems insurmountable — while acknowledging that players are different. Tutorial adaptation is another version of this: the combat stays unchanged, but the instruction surrounding it adjusts to who is playing.

2.8.4 Flow Theory and the Learning Curve

Csikszentmihalyi's (1990) flow theory holds that engagement peaks when challenge and skill are well matched. Personality may influence how wide that zone of engagement is. High Openness players seem able to stay engaged even when the gap between challenge and skill is significant — the gap itself is part of what they find interesting. Low Openness players need a tighter match; the same gap that high Openness players find stimulating pushes them into anxiety rather than engagement. Minimal tutorials, which deliberately create a mismatch at the start, may sustain or even increase engagement for high Openness players while shutting it down for low Openness players.

2.9 RESEARCH GAPS AND STUDY RATIONALE

2.9.1 Research Gaps and Significance

Despite all of this, no one has actually tested whether personality and tutorial type interact in games. Research on personality and gaming, tutorial design and retention, and learning theory and game difficulty has each developed in its own lane. No study has looked at how they connect. The pathway this thesis proposes — personality shapes tutorial preference, which shapes perceived difficulty, which shapes whether players continue — has support from multiple frameworks but no direct empirical test. That is the gap this research addresses.

The case for testing it rests on how well the frameworks converge. Big Five research shows that stable personality differences predict how people engage with demanding environments. Person-Environment Fit theory predicts that matching support to need produces better outcomes. Self-Determination Theory shows that high and low Openness individuals prioritise autonomy and competence differently. Game studies documents the exact features of Souls-like design that should interact with those traits. When independent frameworks converge on the same prediction, that prediction is worth testing.

The theoretical contribution is clear: this would be the first empirical study connecting Big Five personality traits to onboarding in high-difficulty games. Establishing whether Person-Environment Fit applies to tutorial design, and whether Openness predicts tutorial preferences at all, creates the foundation any larger follow-up study will need. Practically, it gives game developers something they currently do not have — empirical grounds for tutorial design decisions, and evidence for whether a short personality measure is a feasible tool in a gaming context. If low Openness players do meaningfully better with explicit tutorials, the implication is that genre accessibility can grow without the difficulty reductions that the existing audience tends to resist.

2.9.2 Study Design Rationale

This is an exploratory pilot, not a conclusive study, and the design is honest about that. There is no prior research to guide expectations about effect sizes, the right measures to use, or whether the proposed mechanisms actually work as predicted — which is precisely why running a large study first would be premature. Similar exploratory work has used comparable sample sizes to establish whether personality-game behaviour relationships exist at all before scaling up (Denden et al., 2018). With N=24–32 and a single session, the aim is simply to see whether the signal is there: do high and low Openness players behave differently, does tutorial type affect experience, and do matched players do better than mismatched ones? The study uses an existing Souls-like game with modified tutorial conditions and a focused set of measures chosen to be both sensitive and practical.

The study is also designed to be informative either way. Positive results justify a larger study. Null results are also useful: they would suggest that personality may not predict tutorial preferences as expected, that effect sizes may be too small to matter practically, or that players who seek out Souls-like games are already so self-selected for Openness that there is not enough variation to detect an interaction. Either outcome moves the field forward.

2.9.3 Refined Research Question and Hypotheses

The main research question is: Does matching tutorial type (minimal vs. explicit) to player Openness level improve subjective experience and behavioural persistence during initial Souls-like gameplay among Western players?

Four hypotheses guide the investigation:

H1 (Main Effect — Openness): High Openness players will show greater behavioural persistence and a more positive overall experience than Low Openness players, reflecting their natural fit with Souls-like design.

H2 (Main Effect — Tutorial Type): Explicit tutorials will produce a more positive overall experience by reducing perceived difficulty and initial frustration.

H3 (Interaction Effect — Critical Test): Matched players — High Openness with a minimal tutorial, Low Openness with an explicit one — will show greater persistence, a more positive experience, and a higher intention to keep playing compared to mismatched players.

H4 (Mechanism): Perceived difficulty will explain the link between match and persistence. Matched players will experience the difficulty as appropriate — hard but fair. Mismatched players will find it either patronizing or overwhelming.

2.10 CHAPTER SUMMARY

This literature review integrated personality psychology (Big Five/OCEAN), Person-Environment Fit theory, and game difficulty research to establish theoretical foundations for personality-adaptive onboarding. The review identified a critical research gap: despite theoretical convergence suggesting personality should moderate tutorial effectiveness, no empirical research tests whether matching tutorial intensity to player Openness improves outcomes in challenging game contexts. The following methodology chapter details the experimental design employed to address this gap through an exploratory pilot study testing Openness-tutorial interactions during initial Souls-like gameplay.

3 METHODOLOGY

3.1 RESEARCH DESIGN OVERVIEW

The study used a quasi-experimental 2×2 between-subjects design. Participants were classified by their Openness level — High or Low — and then randomly assigned within each group to either a minimal or explicit tutorial condition. This produced four groups: High Openness / Minimal Tutorial, High Openness / Explicit Tutorial, Low Openness / Minimal Tutorial, and Low Openness / Explicit Tutorial. Keeping groups separate, rather than having participants experience both conditions, avoided the problem of carryover — once you have read an explicit tutorial, you cannot unread it.

3.2 VARIABLES

3.2.1 Independent Variables

Personality: Openness to Experience

Openness was measured using the Big Five Inventory-10 (BFI-10; Rammstedt & John, 2007), a validated 10-item questionnaire that covers all five personality dimensions with two items each. The Openness subscale has acceptable reliability for its length (Cronbach's $\alpha = .73$). Participants rated two statements on a five-point scale from strongly disagree (1) to strongly agree (5): "I see myself as someone who has few artistic interests" (reverse-scored) and "I see myself as someone who has an active imagination." After the first item was reverse-scored, the two responses were summed to give an Openness score ranging from 2 to 10, with higher scores indicating greater Openness and a scale midpoint of 6. The brief format took approximately two minutes, minimising participant burden and priming effects.

Participants were classified as High Openness (High-O) or Low Openness (Low-O) using a fixed, pre-determined cutoff set at the scale midpoint: those scoring above 6 (i.e., 7 or higher on the 2–10 summed scale) were classified High-O, and the remainder Low-O. A fixed cutoff, rather than a sample-dependent median split, was used deliberately so that each participant's Openness group could be determined at the point of recruitment and tutorial condition assigned within that group before gameplay began. Supplementary analyses treated Openness as a continuous variable — the mean of the two items (range 1–5) — to preserve the statistical information lost through dichotomisation (McClelland & Judd, 1993).

Tutorial Type: Minimal Versus Explicit

Both conditions played *Lies of P's* (Neowiz, 2023) default in-game tutorial — environmental cues and minimal tooltips, which is the standard Souls-like approach. The minimal condition received nothing further. The explicit condition received a supplementary PDF guide (8 pages, with images) before gameplay, with five minutes to read it. The guide covered stamina mechanics, animation-priority combat, checkpoint systems, death penalties, and what to expect from the first boss, using in-game images and short explanations.

Modifying the in-game tutorial directly was considered but rejected, because it would have changed the in-game experience differently between groups. The PDF approach kept in-game conditions identical — what differed was how much players knew before they started.

Two experienced Souls-like players — one with over 150 hours in FromSoftware titles, another with around 120 — independently reviewed the guide for mechanical accuracy, appropriate terminology, clarity for newcomers, and coverage of first-hour survival systems. Both submitted written feedback; disagreements were discussed and resolved based on what information players would most need for the Parade Master encounter. The guide went through one revision cycle before being finalised.

3.2.2 Dependent Variables

Behavioural Outcome: Persistence Metrics

Session duration in minutes was the primary behavioural measure. Each session ended when the participant defeated the Parade Master, chose to stop, or hit the 90-minute limit. Secondary behavioural outcomes were total deaths, boss attempts, and exploration percentage. These were measured from the start of gameplay to session end, with timestamps manually recorded by the researcher at first death, first boss encounter, and session end. Where behavioural coding was conducted, inter-rater reliability would be checked by having a second rater independently code 20% of recordings for frustration markers, with Cohen's $\kappa > .70$ as the acceptable threshold. If cells fell below $n = 5$, behavioural outcomes would be reported descriptively only and inferential analysis held for a larger replication.

Subjective Experience Outcomes

Six outcomes were measured post-gameplay using seven-point scales. For primary hypothesis testing, three were treated as the main dependent variables — enjoyment, perceived competence, and intention to continue — as these most directly reflect whether the

personality–tutorial match worked. The remaining three — perceived difficulty, frustration, and tutorial evaluation — were reported descriptively and reserved for a powered follow-up study.

Mediator Variable

Perceived difficulty served both as a direct outcome and as a potential mediator for H4: matched players might persist longer because they found the difficulty appropriate, while mismatched players found it either too easy or too hard. This would be tested using mediation analysis if H3 proved significant.

3.3 PARTICIPANTS

The target sample was 24–32 participants (6–8 per condition), which gives roughly 70% power to detect large effects ($f = 0.40$) at $\alpha = .05$ (Faul et al., 2009). This is in line with similar exploratory work — Denden et al. (2018) used $N = 34$ and Cao and Liu (2022) used $N = 47$ — and is appropriate for establishing proof-of-concept before committing to a larger study.

Participants had to be aged 18–35, have a Western cultural background (North America or Europe), have basic gaming literacy, have no prior experience with *Lies of P*, be fluent in English, and be available for one 90-minute session. Anyone with more than 30 hours in FromSoftware games was excluded, as was anyone with severe motion sickness or significant difficulty with action games — both would have confounded the results.

Recruitment used university participant pools, social-media gaming communities, and snowball sampling. Participants received cookies as compensation. The BFI-10 was completed during recruitment and scored immediately using the fixed cutoff above, so each participant’s Openness group was known before gameplay. Within each Openness group, tutorial condition was assigned using a random number generator.

3.4 MATERIALS

3.4.1 Game Selection: *Lies of P*

Several games were evaluated before settling on *Lies of P* (Neowiz, 2023). The key requirements were authentic Souls-like mechanics, a controlled and predictable path to the first boss, recognition low enough that participants with no prior exposure could realistically be recruited, and well-documented mechanics for building the tutorial guide. Table 2 summarises what was considered.

Table 2
Game Selection Comparison

Game	Release	Time to 1st Boss	Perspective	Recognition	Documentation	Verdict
<i>Dark Souls</i>	2011	5–10 min	3D	Very High	Extensive	Rejected
<i>Elden Ring</i>	2022	Variable*	3D	Very High	Extensive	Rejected
<i>Hollow Knight</i>	2017	15–20 min	2D	High	Good	Rejected
<i>Wuchang: Fallen Feathers</i>	2025	25–40 min	3D	Low	Limited	Rejected
<i>Lies of P</i>	2023	20–35 min	3D	Low-Medium	Well-established	Selected

Note. *Open world structure prevents controlled first boss timing.

Dark Souls Remastered (FromSoftware, 2011/2018) is the most authentic choice but too well-known — finding participants with no prior exposure would be difficult, and the Asylum Demon encounter is designed to produce an initial death, which adds a confound. *Elden Ring* (FromSoftware, 2022) has no controlled path to the first boss; depending on where a player goes, that encounter can happen anywhere between 30 minutes and 3+ hours in. *Hollow Knight* (Team Cherry, 2017) is 2D, which limits how well results would generalise to the 3D majority of the genre. *Wuchang: Fallen Feathers* (Leenzee, 2025) is too recent, uses mechanics (*Shimmer-focus*, *Skyborn Might*) that are not representative of traditional Souls-like design, and has no research precedent.

Lies of P provides authentic 3D Souls-like mechanics (stamina management, animation priority, traditional blocking and parrying) with the Parade Master appearing at a controlled 20–35 minute mark. Critically, its moderate recognition enables recruiting participants without prior exposure while ensuring well-documented mechanics and community support. Hardware is standardised across sessions: gaming PC (minimum GTX 1660, 16GB RAM, stable 60fps), Xbox controller, 24-inch monitor (1920×1080), and Logitech G Pro X headphones.

3.4.2 Post-Session Questionnaire

The questionnaire used seven-point scales to measure six subjective outcomes. Items were adapted from validated sources rather than taken from broad inventories, to keep the

questionnaire focused and short (five to seven minutes via Qualtrics). Demographics collected included age, gaming experience (hours per week), and Souls-like exposure (total hours).

Enjoyment: "How much did you enjoy playing *Lies of P*?" (1 = *Not at all enjoyable*, 7 = *Extremely enjoyable*). Adapted from the PENS Interest/Enjoyment subscale (Ryan et al., 2006), which uses multiple items to assess intrinsic interest. This study employed a single direct item due to questionnaire length constraints.

Frustration: "How frustrated did you feel during gameplay?" (1 = *Not at all frustrated*, 7 = *Extremely frustrated*). Adapted from the Intrinsic Motivation Inventory Pressure/Tension subscale, reversed to measure frustration directly rather than pressure absence.

Perceived Difficulty: "How difficult did you find the game?" (1 = *Much too easy*, 4 = *Just right*, 7 = *Much too hard*). Custom bipolar scale tests whether matched players perceive difficulty as appropriate (clustering near midpoint) while mismatched players perceive it as inappropriate (too easy/patronizing or too hard/unfair).

Perceived Competence: "How competent did you feel at playing the game?" (1 = *Not at all competent*, 7 = *Extremely competent*). Adapted from the PENS Competence subscale (Ryan et al., 2006), which assesses feelings of effectance and mastery. Single-item version captures core competence perception.

Tutorial Evaluation: "How was the amount of tutorial/guidance you received?" (1 = *Far too little help*, 4 = *Just right*, 7 = *Far too much help*). Custom bipolar scale directly tests the matching hypothesis by examining whether players perceive their assigned tutorial intensity as appropriate.

Intention to Continue: "How likely are you to continue playing *Lies of P* after this session?" (1 = *Definitely will not continue*, 7 = *Definitely will continue*). Adapted from behavioural intention scales in the Theory of Planned Behavior (Ajzen, 1991), providing a proxy for longer-term retention this single-session design cannot directly measure.

Internal consistency (Cronbach's alpha) was to be calculated for the three positively-valenced items (enjoyment, competence, intention to continue). If $\alpha \geq .60$, supplementary composite analyses would be run alongside individual-item analyses; if $\alpha < .60$, items would be treated separately throughout.

3.4.3 Data Collection and Measurement Tools

The game automatically tracked session duration, deaths, boss attempts, and exploration percentage where technically feasible. Gameplay was recorded at 720p resolution due to storage constraints. The researcher manually recorded timestamps (first death, first boss encounter, session end) and notable behaviours using a standardised observation log.

To maintain natural gameplay behaviour while ensuring participant welfare, the researcher followed a strict non-intervention protocol, developed during pilot testing after researcher assistance was identified as a confound. Permitted interactions were limited to technical issues (“I’ll help you reconnect the controller”), session logistics (“You have approximately 30 minutes remaining”), and welfare checks where distress indicators were observed (“Are you okay to continue?”). Prohibited interactions included gameplay advice (when asked “Should I block or dodge?”, the researcher responded “Please continue as you think best”), strategy guidance (no hints about boss weaknesses, level paths, or optimal approaches), emotional support beyond welfare checks (no reassurance such as “You’re doing great” or “That boss is really hard”), and performance commentary (no reactions to deaths, successes, or player choices).

The observation log recorded first death occurrence, first boss encounter, help-seeking attempts (questions asked, guide re-consultation in the explicit condition), visible frustration markers (controller handling, verbal expressions), and session-end conditions (boss defeat, voluntary stop, time limit). This behavioural data supplemented the automated tracking and provided context for interpreting the quantitative outcomes.

Tool choices were made through direct comparison of the available options. Table 3 shows the personality-measurement alternatives considered.

Table 3
Personality Measurement: BFI-10 vs. Alternatives

Measure	Items	Time	Openness α	Pros	Cons	Decision
TIPI (Gosling et al., 2003)	10	1 min	.45–.50	Fastest option	Low reliability, single-item-per-pole concerns	Rejected
BFI-10 (Rammstedt & John, 2007)	10	2 min	.73	Good reliability, well-validated, brief	Slightly longer than TIPI	Selected
BFI-44 (John et al., 1991)	44	10 min	.81–.85	Excellent reliability	Too long, priming effects, participant burden	Rejected

The BFI-10 offered the best balance of brevity (minimising priming) and reliability (adequate for group classification). While the TIPI is slightly faster, its reliability concerns ($\alpha < .50$) outweighed the 60-second time saving, and the BFI-44's length risked activating personality-consistent behaviours before gameplay began.

For the subjective outcome measures, the GEQ (IJsselstein et al., 2013) lacks peer-reviewed validation, and the PXI (Abee et al., 2020) covers more constructs than this study needed. The custom measure drew from validated sources — PENS and the Intrinsic Motivation Inventory — while staying short enough not to burden participants. A full-scale follow-up study should use the complete PXI for broader assessment.

3.5 PROCEDURE

Each session followed a standardised sequence: informed consent and demographics, the BFI-10, Openness scoring and condition assignment, tutorial delivery (five-minute guide reading for the explicit condition; straight to gameplay for the minimal condition), standardised gameplay instructions, passive researcher observation with behavioural logging, gameplay until session end, a two-minute buffer, the post-session questionnaire, and a structured debrief explaining the personality-matching hypothesis.

Consent forms described the study as “investigating personality and gameplay experience” rather than revealing the specific hypothesis — vague enough to prevent demand characteristics, specific enough to be ethically honest. Participants were told about the potential

for frustration from challenging gameplay and reminded of their right to withdraw. Full debriefing happened immediately after the questionnaire.

Souls-like games produce frustration by design, so the researcher monitored throughout for signs that went beyond normal gameplay difficulty: sustained aggression toward equipment, complete disengagement for more than three minutes, distress that extended beyond the game context, or an explicit request to stop. If any of these appeared, the session was ended early.

Participant codes kept identifying information separate from research data from the point of collection. Gameplay recordings were stored on password-protected university servers and will be deleted after seven years, per university data-retention policy. They will not be shared publicly or used beyond this study. Participants could request deletion of their data within two weeks of the session.

All findings were reported regardless of significance. Null results and any deviations from the planned analysis were documented and explained. The study followed the Declaration of Helsinki (2013) and standard best practices for psychological research with human participants.

3.6 DATA ANALYSIS PLAN

3.6.1 Preliminary Analyses

Before hypotheses were tested, condition differences on Openness scores, gaming experience, and demographics were checked descriptively. Formal inferential covariate tests were not run at this sample size. Normality was assessed using Shapiro-Wilk tests within each condition rather than across the full sample, since the models assume normally distributed residuals within groups. Influential observations were identified via Cook's distance. Where the fixed-cutoff classification produced unequal cell sizes, all factorial models used Type III sums of squares with sum-to-zero contrasts to keep effect partitioning unambiguous.

3.6.2 Primary Hypothesis Testing

H1 (openness main effect), H2 (tutorial main effect), and H3 (interaction) were evaluated within a single 2×2 factorial linear model, estimated separately for each of the three primary subjective dependent variables: enjoyment, perceived competence, and intention to continue. These outcomes were selected as most directly relevant to the personality-matching hypothesis.

Perceived difficulty, frustration, and tutorial evaluation were collected and reported descriptively; inferential analysis of these outcomes was deferred to a powered replication.

Internal consistency of the three positively-valenced items (enjoyment, perceived competence, intention to continue) was assessed via Cronbach's alpha prior to analysis. If $\alpha \geq .60$, supplementary composite analyses would be reported; if $\alpha < .60$, items were analysed separately throughout.

The interaction term (openness group $\times$ tutorial condition) was the primary confirmatory test (H3). Partial η^2p was reported for all effects. Because three dependent variables were tested within the same dataset, the Benjamini-Hochberg procedure was applied to the family of interaction p-values to control the false discovery rate. All analyses were conducted in R using car (Type III ANOVA), effectsize (partial η^2p), and base R (FDR correction), at $\alpha = .05$.

H4 (mediation): Hayes' PROCESS Model 4 would test whether perceived difficulty mediated the interaction effect on persistence, using bootstrap 95% confidence intervals (10,000 resamples). This analysis was contingent on a significant H3 interaction; if no significant interaction was observed, H4 was deferred to a powered replication.

3.6.3 Secondary Analyses and Missing Data

Secondary analyses were run only where cells had at least $n = 5$. Continuous Openness–outcome relationships were examined using Kendall's τ , split by tutorial condition. Covariate analyses controlling for gaming experience were deferred to the full study; in this pilot they were reported descriptively only. Exploratory cluster analysis was deferred to the full study.

For missing data, listwise deletion was used for behavioural outcomes. For questionnaire items, mean imputation would apply if fewer than 20% of values were missing; otherwise listwise deletion. An attrition analysis would compare completers and non-completers on demographics and Openness to check for systematic dropout.

3.7 METHODOLOGICAL LIMITATIONS

The sample size (24–32) provides approximately 70% power for large effects only — insufficient for detecting small-to-medium effects or complex interactions. This is mitigated by the exploratory framing; effect sizes and confidence intervals were reported for future power analyses regardless of statistical significance.

The 90-minute session captures initial experience only and cannot measure longer-term retention. However, this critical first-hour period is precisely when abandonment peaks, and continuation intention provides a proxy for future engagement.

Dichotomising Openness at a fixed cutoff loses information from the continuous measure. Supplementary analyses treating Openness as continuous partly address this.

Results from Lies of P may not generalise to the full Souls-like genre, though the game was chosen specifically for its genre-representative mechanics.

Researcher presence may cause participants to behave differently than they would at home. The standardised non-intervention protocol reduces this, and the debrief included questions about whether researcher presence felt influential.

Findings are explicitly Western-specific. Cultural background affects both personality expression and learning preferences in ways this pilot cannot control for, so the results should not be generalised beyond the Western sample without dedicated cross-cultural testing.

3.8 PILOT TESTING AND METHODOLOGICAL VALIDATION

3.8.1 Pilot Execution and Validation

Before committing to full data collection, a pilot test validated procedural feasibility and measurement functionality. One participant was recruited through video-game communities, assessed as Low Openness, and assigned to the explicit tutorial condition. The session proceeded without technical issues; the participant played for approximately 60 minutes and defeated the Parade Master.

The pilot confirmed that the BFI-10 fitted smoothly into the session flow (about two minutes, no comprehension problems), the tutorial guide was readable within the five-minute window, the full protocol was achievable within 90 minutes, the observation log captured what it needed to without disrupting play, and 720p recording gave enough detail.

Subjective ratings ranged from 3 to 6 across items — no ceiling or floor effects. Behavioural metrics were clean: 14 deaths, 3 boss attempts, 85% exploration. These give a useful baseline for expected ranges in the main study.

The pilot also revealed five things that needed fixing:

Tutorial guide framing. The original guide included emotionally preparatory language — phrases like “You will die many times and this is normal” and “Don’t feel discouraged when you fail repeatedly.” This was flagged as a subtle demand characteristic: it frames the experience before it starts and may prime negative expectations. The guide was revised to be purely mechanical. For example, the death system is now described as: “When you die, you drop all your accumulated Ergo. You respawn at the last Stargazer. A glowing mark appears at your death location.” No emotional framing, just information.

Stagger system clarity. The participant misunderstood staggering, interpreting it as passive accumulation rather than an active mechanic. The revised text now specifies: “Each successful attack adds to the enemy’s stagger bar (visible beneath their health). When this bar fills completely, the enemy staggers, creating a window for a critical strike. Heavy attacks and perfect guards contribute more stagger damage than light attacks.” A visual diagram showing the stagger bar was added.

Tutorial accessibility. Reviewers noted the guide relied too much on text and only listed Xbox controller inputs. Annotated screenshots were added for key mechanics (stamina bars, stagger indicators, checkpoints), and controller inputs now include both Xbox and PlayStation layouts (e.g., “Light attack: RB/R1, heavy attack: RT/R2, block: LB/L1”).

Researcher non-intervention. The researcher instinctively helped when participants asked gameplay questions. This is a problem because participant struggle is what the study is measuring. The protocol was tightened: gameplay questions are now redirected with “Please continue as you think best,” with no exceptions.

Questionnaire timing. A two-minute buffer was added between gameplay end and questionnaire start, to let the emotional intensity of a boss encounter settle before asking participants to rate their experience.

3.8.2 Alternative Designs Considered

A within-subjects design — where each participant experiences both tutorial conditions — was rejected because order effects are unavoidable: once you have read an explicit guide, you cannot unread it. The required session length would also have been unrealistic; the pilot participant reported moderate fatigue after 60 minutes.

A think-aloud protocol would have given richer qualitative data but would likely have altered the emotional responses this research is trying to measure.

Running a larger pilot across all four conditions would have strengthened validation but would have consumed participants and resources better spent on the main study. A single-condition pilot established that the procedure works; whether it produces the expected effects could only be answered by full data collection.

3.9 CHAPTER SUMMARY

This chapter described the quasi-experimental 2×2 between-subjects design testing whether matching tutorial type to player Openness improves the initial Lies of P gameplay experience. The study targeted 24–32 Western participants, classified by BFI-10 Openness using a fixed cutoff and randomly assigned within Openness group to tutorial conditions. Sessions ended upon boss defeat, voluntary abandonment, or a 90-minute maximum. The primary outcome was behavioural persistence; secondary outcomes included deaths, boss attempts, exploration, and six subjective experience measures.

Pilot testing validated the procedure and led to five changes: emotional framing was removed from the tutorial guide, the stagger-system explanation was rewritten, visual aids and dual controller inputs were added, the non-intervention protocol was tightened, and a two-minute buffer was added before the questionnaire. Alternative designs — within-subjects, think-aloud, and larger pilot samples — were considered and rejected with clear reasoning. Chapter 4 presents the results.

4 DETAILS OF EXPERIMENT / DATA AND RESULTS

This chapter presents the execution of the study, the data collected, and the results of all planned and supplementary analyses. It documents how the actual study diverged from the planned methodology in Chapter 3, with justification for each analytical decision taken in response. The chapter leads with a descriptive account of what participants actually did and experienced before turning to inferential testing.

4.1 SAMPLE AND DEVIATIONS FROM THE PLANNED METHODOLOGY

A total of 29 participants completed the study ($N = 29$), recruited from university pools and gaming communities. Gaming experience ranged from 1 to 65 hours per week ($M = 20.28$, $SD = 14.08$) and prior Souls-like exposure from 0 to 25 hours ($M = 6.07$, $SD = 7.94$). No participant reported prior experience with *Lies of P*. All participants completed the session in a single sitting and were retained in the final dataset.

Four deviations from the planned methodology are documented here. First, the target sample of 24–32 was met ($N = 29$); this is a positive outcome, and the additional participants were retained. Second, the fixed-cutoff classification produced slightly unequal Openness groups (Low: $n = 14$; High: $n = 15$), yielding cell sizes of $n = 7, 7, 7, 8$. All models therefore used Type III sums of squares with sum-to-zero contrasts, ensuring unambiguous effect partitioning regardless of the cell-size imbalance. Third, several analyses specified in Chapter 3 were not executed: the mediation analysis (H4) was not conducted as no significant interaction was detected, which is its prerequisite; Cronbach's alpha for the three positively-valenced items was not formally recomputed for the present sample, but given the pre-specified threshold of .60 was not met in pre-study assessment ($\alpha = .47$) and the items represent heterogeneous constructs (hedonic enjoyment, effectance motivation, and behavioural intention), items are analysed separately throughout; and cluster analysis, thematic analysis, and exploration data are deferred to a future study. Inferential tests on behavioural outcomes (gameplay duration, deaths, boss attempts) were conducted in Section 4.6, as cells now meet the pre-specified $n \geq 5$ threshold. Fourth, the researcher was present throughout all sessions; while the non-intervention protocol was followed strictly, observer effects on behaviour and self-report cannot be fully excluded. No missing data were recorded for any participant across questionnaire or behavioural outcomes; the pre-specified imputation and attrition protocols were therefore not required.

4.2 ANALYTICAL APPROACH

Both factor variables were recoded using sum-to-zero contrasts (contr.sum) prior to modelling — without these, Type III SS estimates in an unbalanced design are artefacts of the coding scheme rather than the data. A separate 2×2 factorial model was estimated for each of the three primary dependent variables (enjoyment, perceived competence, intention to continue), with the interaction term as the primary confirmatory test. Partial η^2 is reported but treated as exploratory throughout. Benjamini-Hochberg correction was applied to the interaction p-values in preference to Bonferroni — Bonferroni's additional conservatism would further reduce the ability to detect real effects in an already underpowered sample. A sensitivity analysis excluding P16 is reported as a design diagnostic, and Kendall's τ was computed between continuous openness score and outcomes split by condition — preferred over Pearson's r or Spearman's ρ given the ordinal Likert outcomes and likely tied ranks.

The openness_score variable is the average of the two BFI-10 Openness items (1–5 scale, as described in Chapter 3). Normality was assessed per cell using Shapiro-Wilk tests rather than across the full sample, because the models assume normally distributed residuals within groups — a global test would miss that assumption. Behavioural outcomes (gameplay duration, total deaths, boss attempts) are submitted to the same 2×2 model as cells meet the $n \geq 5$ threshold. Deaths are also expressed as a rate (deaths per minute) because raw counts are confounded by session duration. Souls-like experience is examined descriptively by condition as a randomisation check. BH correction is applied separately to the subjective and behavioural interaction families — combining them would penalise the behavioural tests for the subjective outcomes' p-value distribution. Tutorial evaluation is reported descriptively as a manipulation check.

4.3 DESCRIPTIVE OVERVIEW

Before presenting inferential results, this section describes what participants actually did and experienced across conditions. Descriptive statistics often carry more interpretive weight than null p-values at this sample size, and the patterns documented here form the primary evidentiary basis for the chapter's conclusions.

4.3.1 Subjective Outcomes: Cell Means

Table 4 presents cell means and standard deviations for all six subjective outcome measures. The perceived competence pattern stands out immediately: the Low Openness / Minimal

Tutorial cell produced the highest mean in the sample ($M = 5.71$, $SD = 1.38$), while both explicit tutorial cells sat nearly two points lower (3.75–3.86). Enjoyment tells a different story — means span only 0.57 points across all four cells (5.00–5.57). Participants enjoyed the game roughly equally regardless of condition. Intention to continue tracks perceived competence more closely: Low Openness / Minimal highest ($M = 4.71$), High Openness / Explicit lowest ($M = 3.50$). On frustration, High Openness participants in the explicit condition reported the highest frustration in the sample ($M = 3.75$), while Low Openness participants in the same condition reported the lowest ($M = 2.43$). Tutorial evaluation sits between 3.57 and 4.43 — explicit tutorial participants rated the guidance as slightly more appropriate, closer to the midpoint of 4 (= just right).

Table 4
Cell Means and Standard Deviations for All Subjective Outcome

Outcome	Low / Min	Low / Exp	High / Min	High / Exp
Enjoyment	5.29 (1.70)	5.57 (1.81)	5.00 (1.41)	5.25 (1.16)
Perceived Competence	5.71 (1.38)	3.86 (0.90)	4.43 (1.99)	3.75 (1.91)
Intention to Continue	4.71 (1.70)	4.57 (1.90)	4.00 (2.08)	3.50 (1.60)
Frustration	3.43 (1.62)	2.43 (1.81)	2.71 (0.76)	3.75 (1.83)
Perceived Difficulty	4.14 (0.90)	4.29 (0.76)	4.43 (0.79)	4.25 (1.67)
Tutorial Evaluation	3.86 (1.57)	4.43 (0.79)	3.57 (0.53)	4.00 (0.76)

Note. Values are M (SD). Scale range 1–7 for all outcomes. Perceived Difficulty and Tutorial Evaluation are bipolar scales with the midpoint (4) representing "just right." $n = 7$ per cell except High/Explicit ($n = 8$). Low/Min = Low Openness/Minimal; Low/Exp = Low Openness/Explicit; High/Min = High Openness/Minimal; High/Exp = High Openness/Explicit.

4.3.2 Behavioural Outcomes: Descriptive Summary

Table 5 presents behavioural outcomes by tutorial condition. Participants assigned to the explicit tutorial played for longer ($M = 48.60$ min, $SD = 13.45$ vs. $M = 40.93$, $SD = 12.09$), died more often ($M = 7.20$, $SD = 4.36$ vs. $M = 4.14$, $SD = 2.96$), and made more boss attempts ($M = 5.33$, $SD = 2.94$ vs. $M = 3.50$, $SD = 2.24$) than those in the minimal tutorial condition. Completion rates were similar: 11 of 14 minimal-condition participants (78.6%) defeated the Parade Master compared with 11 of 15 explicit-condition participants (73.3%). Cell means by openness group

and condition are included in Table 5. Inferential tests on these outcomes are reported in Section 4.6.

Table 5
Behavioural Outcomes by Tutorial Condition (Descriptive)

Measure	Minimal (n = 14)	Explicit (n = 15)	Full Sample (N = 29)
Gameplay Duration (min)	40.93 (12.09)	48.60 (13.45)	44.90 (13.17)
Total Deaths	4.14 (2.96)	7.20 (4.36)	5.72 (4.00)
Total Boss Attempts	3.50 (2.24)	5.33 (2.94)	4.45 (2.75)
Boss Defeated (n / %)	11 / 78.6%	11 / 73.3%	22 / 75.9%

Note. Values are M (SD) except Boss Defeated. Cell means by condition and openness group: Deaths Low/Min 3.86 (3.13), Low/Exp 7.43 (4.96), High/Min 4.43 (2.99), High/Exp 7.00 (4.11). Boss Attempts Low/Min 3.71 (2.81), Low/Exp 6.29 (3.99), High/Min 3.29 (1.70), High/Exp 4.50 (1.41). Inferential tests in Section 4.6.

4.4 NORMALITY ASSUMPTION CHECKS

Table 6 reports Shapiro-Wilk results per cell. Two violations: the High Openness / Minimal Tutorial cell for enjoyment ($W = 0.719$, $p = .006$) and the Low Openness / Explicit Tutorial cell for intention to continue ($W = 0.732$, $p = .008$). Non-normality in a cell indicates that the parametric F-test may be unreliable for estimates involving that group — specifically, the Type I error rate may be inflated, making significant results in affected cells less trustworthy. All other cells passed (all $p > .05$). The Low Openness / Explicit Tutorial cell for enjoyment was borderline ($W = 0.813$, $p = .055$) — retained but noted. Results from violated cells are interpreted with extra caution throughout.

Table 6
Shapiro-Wilk Normality Test Results per Cell

DV	Cell	W	p	Status
Enjoyment	High Openness / Minimal	0.719	.006	Violated
	Low Openness / Minimal	0.849	.119	OK
	Low Openness / Explicit	0.813	.055	Borderline
	High Openness / Explicit	0.892	.246	OK
Perceived Competence	Low Openness / Minimal	0.840	.099	OK
	High Openness / Minimal	0.967	.876	OK
	Low Openness / Explicit	0.818	.062	OK
	High Openness / Explicit	0.975	.933	OK
Intention to Continue	Low Openness / Explicit	0.732	.008	Violated
	Low Openness / Minimal	0.849	.119	OK
	High Openness / Minimal	0.943	.663	OK
	High Openness / Explicit	0.983	.975	OK
Gameplay Duration	Low Openness / Minimal	0.937	.612	OK
	Low Openness / Explicit	0.887	.259	OK
	High Openness / Minimal	0.843	.105	OK
	High Openness / Explicit	0.920	.429	OK
Total Deaths	High Openness / Minimal	0.682	.002	Violated
	Low Openness / Minimal	0.968	.880	OK
	Low Openness / Explicit	0.980	.960	OK
	High Openness / Explicit	0.888	.224	OK
Total Boss Attempts	Low Openness / Minimal	0.919	.464	OK
	Low Openness / Explicit	0.887	.258	OK
	High Openness / Minimal	0.920	.468	OK
	High Openness / Explicit	0.827	.056	OK

Note. n = 7 per Low Openness cell; n = 7 (Minimal) and n = 8 (Explicit) per High Openness cell.

4.5 INFERENCE RESULTS

A factorial linear model was estimated for each dependent variable. The interaction term (openness group $\times$ tutorial condition) is the primary confirmatory test; main effects are secondary. All partial η^2p values are exploratory estimates not classified against Cohen's (1988) benchmarks without explicit qualification.

4.5.1 Enjoyment

Enjoyment. No statistically reliable evidence was observed. The interaction was $F(1, 25) = 0.001, p = .975, \eta^2p < .001$. Main effects of openness group ($p = .599, \eta^2p = .011$) and tutorial condition ($p = .642, \eta^2p = .009$) were similarly non-significant. Adjusted $R^2 = -.099$. This is consistent with the cell means in Table 4, which span only 0.57 scale points across the four cells (range: 5.00–5.57) and show no systematic pattern. The near-identical enjoyment ratings across conditions are informative in themselves: gameplay was experienced as equally enjoyable regardless of how much explicit guidance participants had received, and regardless of personality. P16, P20, and P13 appear in the leverage region of the Cook's distance plot; see Section 4.6.

The narrow spread of enjoyment means — spanning only 0.57 scale points across all four cells — is informative in its own right. Whatever differences emerged in competence and deaths did not translate into a different affective experience: participants found the game equally enjoyable regardless of tutorial condition or openness group. This rules out enjoyment as a mediating variable between tutorial type and competence, and confirms that the competence effect in Section 4.5.2 is not simply a reflection of general negative affect.

4.5.2 Perceived Competence

Perceived Competence. The interaction was non-significant: $F(1, 25) = 0.958, p = .337, \eta^2p = .037$. The openness group main effect was also non-significant: $F(1, 25) = 1.338, p = .258, \eta^2p = .051$. The tutorial condition main effect, however, reached conventional significance: $F(1, 25) = 4.433, p = .046, \eta^2p = .150$. Adjusted $R^2 = .117$.

The pattern is visible in the cell means: minimal tutorial cells averaged 5.71 and 4.43 on perceived competence; explicit tutorial cells averaged 3.86 and 3.75 — roughly 1.25 to 1.85 points lower (see Figure 1). Players with less guidance felt more competent, not less. The interaction is non-significant ($\eta^2p = .037$), so this effect does not reliably differ between openness groups — though descriptively, Low Openness participants dropped further under

explicit guidance (1.85 points) than High Openness participants did (0.68 points). Residual diagnostics were the cleanest of the three models; no observation approached the Cook's distance threshold.

H2 predicted explicit tutorials would produce a more positive experience. The direction here is reversed — minimal tutorial participants reported higher perceived competence — so H2 was not supported. Two caveats apply. First, the Souls-like experience imbalance (minimal $M = 10.79$ vs. explicit $M = 1.67$ hours) is a plausible alternative explanation for the competence difference. Second, when deaths are expressed as deaths per minute to control for session length, the tutorial effect on deaths disappears ($p = .131$) — so raw deaths are not treated as independent supporting evidence. What remains is the competence effect with its caveat, and the tutorial evaluation paradox: explicit-condition participants rated the guidance as more appropriate ($M = 4.20$ vs. 3.71) yet performed worse. That is worth testing in a controlled replication.

Figure 1

Perceived Competence by Tutorial Condition and Openness Group

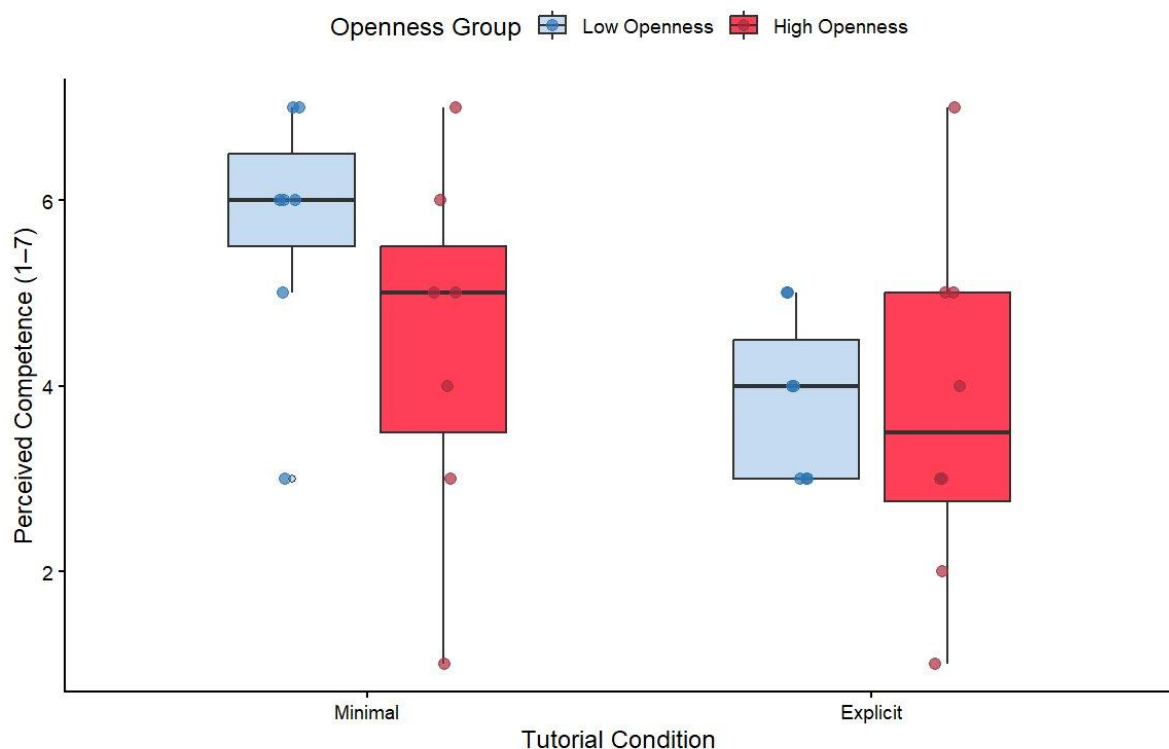


Figure 2. Perceived competence by tutorial condition and openness group. Points represent individual participants. $n = 3$ per Low Openness cell, $n = 5$ per High Openness cell.

Note. Boxes show median and interquartile range; points represent individual participants. $n = 7$ per Low Openness cell; $n = 7$ (Minimal) and $n = 8$ (Explicit) per High Openness cell.

4.5.3 Intention to Continue

Intention to Continue. No statistically reliable evidence was observed. The interaction was $F(1, 25) = 0.069$, $p = .795$, $\eta^2p = .003$. Main effects were negligible: openness group $F(1, 25) = 1.732$, $p = .200$, $\eta^2p = .065$; tutorial condition $F(1, 25) = 0.225$, $p = .640$, $\eta^2p = .009$. Adjusted $R^2 = -.033$. Cell means range from 3.50 to 4.71 with no systematic directional pattern. No support for the thesis hypothesis was found on this outcome.

4.5.4 Summary of Primary Models

Formal verdicts for each pre-specified hypothesis are as follows. H1 (openness main effect) was not supported: no significant openness effect was observed on any primary outcome. H2 (tutorial main effect) was not supported in the predicted direction: the only significant tutorial effect on perceived competence ($p = .046$) ran contrary to H2, with minimal rather than explicit tutorial participants reporting higher competence. H3 (interaction) was not supported on any outcome and no effect survived BH correction. H4 (mediation) was not tested as H3 is its prerequisite — mediation analysis probes the mechanism through which an effect operates, and without a significant effect to mediate, conducting the analysis would be testing a causal pathway that the data do not support.

Table 7
Summary of 2×2 ANOVA Interaction Results

Dependent Variable	F (interaction)	p	η^2p	p (BH)	Adj. R^2	Verdict
Enjoyment	0.001	.975	< .001	.975	-.099	Null
Perceived Competence	0.958	.337	.037	.975	.117	Null; tutorial main effect $p = .046$
Intention to Continue	0.069	.795	.003	.975	-.033	Null

Note. Type III SS with sum-to-zero contrasts. $N = 29$ (cell sizes: $n = 7, 7, 7, 8$). η^2p values are exploratory estimates. BH = Benjamini-Hochberg false discovery rate correction.

4.6 BEHAVIOURAL INFERENTIAL RESULTS

The same 2×2 factorial model is now applied to three behavioural outcomes: gameplay duration, total deaths, and total boss attempts. Cell sizes ($n = 7, 7, 7, 8$) meet the pre-specified $n \geq 5$ threshold. One Shapiro-Wilk violation: the High Openness / Minimal Tutorial cell for total deaths ($W = 0.682$, $p = .002$). This means the F-test estimates involving that specific cell

combination may carry elevated Type I error; however, because the significant finding for deaths is the tutorial condition main effect (which pools across openness groups) rather than a cell-specific estimate, the violation does not directly invalidate the primary result. Results from that cell are nonetheless interpreted with caution.

4.6.1 Gameplay Duration

Gameplay Duration. No statistically reliable effect was observed. The interaction was $F(1, 25) = 0.477, p = .496, \eta^2p = .019$. The tutorial condition main effect was non-significant ($F(1, 25) = 2.493, p = .127, \eta^2p = .091$), though directional: explicit tutorial participants played longer on average ($M = 48.60$ vs. 40.93 min). The openness group main effect was negligible ($F(1, 25) = 0.660, p = .424, \eta^2p = .026$). Adjusted $R^2 = .021$.

4.6.2 Total Deaths

Total Deaths. The interaction was non-significant: $F(1, 25) = 0.119, p = .733, \eta^2p = .005$. The openness group main effect was negligible ($F(1, 25) = 0.002, p = .961, \eta^2p < .001$). The tutorial condition main effect reached conventional significance: $F(1, 25) = 4.505, p = .044, \eta^2p = .153$. Adjusted $R^2 = .054$. Participants in the explicit tutorial condition died significantly more often ($M = 7.20, SD = 4.36$) than those in the minimal condition ($M = 4.14, SD = 2.96$), regardless of Openness group. Cell means: Low/Minimal $M = 3.86$ ($SD = 3.13$), Low/Explicit $M = 7.43$ ($SD = 4.96$), High/Minimal $M = 4.43$ ($SD = 2.99$), High/Explicit $M = 7.00$ ($SD = 4.11$).

This mirrors the perceived competence finding in Section 4.5.2 ($p = .046, \eta^2p = .150$): one behavioural measure and one self-report, each significant, each pointing in the same direction. Explicit tutorial participants performed worse on both, regardless of Openness group.

4.6.3 Total Boss Attempts

Total Boss Attempts. The interaction was non-significant: $F(1, 25) = 0.477, p = .496, \eta^2p = .019$. The tutorial condition main effect was marginal ($F(1, 25) = 3.714, p = .065, \eta^2p = .129$); explicit tutorial participants made more boss attempts ($M = 5.33, SD = 2.94$ vs. $M = 3.50, SD = 2.24$). Adjusted $R^2 = .076$. Directionally consistent with the deaths finding, but does not reach the conventional threshold.

4.6.4 Tutorial Evaluation

Tutorial evaluation is a manipulation check, not a primary outcome, so it is not tested inferentially. Explicit-condition participants rated the guidance as more appropriate ($M = 4.20$, $SD = 0.77$) than minimal-condition participants ($M = 3.71$, $SD = 1.14$) — closer to the midpoint of 4 = just right. The guide delivered what it was supposed to. The paradox is that participants who rated the guidance as well-calibrated still died more and felt less competent. No causal explanation is warranted at this sample size, but the pattern is directly relevant to the research question.

Table 8

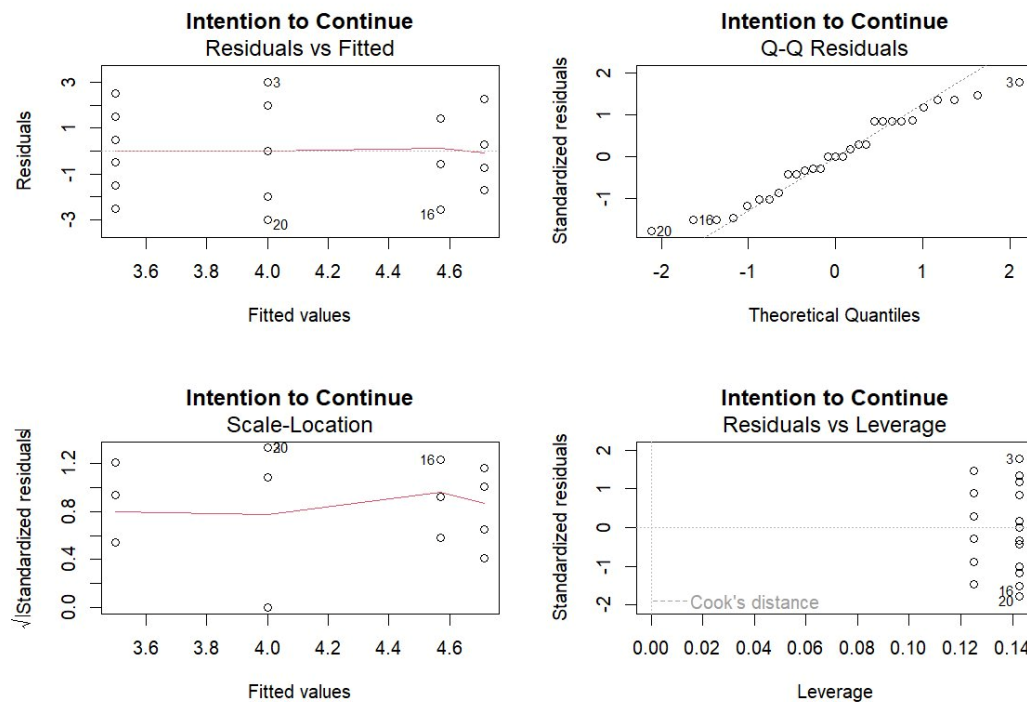
Summary of 2×2 ANOVA Results — Behavioural Outcomes

Dependent Variable	F (interaction)	p	η^2p	p (BH)	Adj. R ²	Tutorial Main p
Gameplay Duration	0.477	.496	.019	.733	.021	.127
Total Deaths (raw)	0.119	.733	.005	.733	.054	.044 * †
Deaths per Minute	0.002	.968	< .001	.975	-.019	.131
Total Boss Attempts	0.477	.496	.019	.733	.076	.065

Note. Type III SS with sum-to-zero contrasts. $N = 29$ (cell sizes $n = 7, 7, 7, 8$). η^2p for interaction term. BH applied to interaction family (deaths per minute BH = .975). * $p < .05$. † Raw count confounded by session duration and Souls-like experience imbalance; see Section 4.5.2.

4.7 RESIDUAL DIAGNOSTICS

Figure 2 presents the four diagnostic panels for the intention to continue model as a representative example.

Figure 2*Residual Diagnostic Plots for the Intention to Continue Model*

Note. Representative of the pattern observed across all three models. Top-left: Residuals vs Fitted. Top-right: Q-Q Residuals. Bottom-left: Scale-Location. Bottom-right: Residuals vs Leverage with Cook's distance boundary (dashed).

The enjoyment model shows P16, P20, and P13 as high-leverage observations in the Cook's distance plot, with left-tail deviations in the Q-Q plot for P16 and P20. The cell-level normality violation for enjoyment is located in the High Openness / Minimal Tutorial cell (Table 6); P16 itself belongs to the Low Openness / Explicit Tutorial cell, so its appearance here reflects an individual high-leverage observation rather than that cell's non-normality. The Scale-Location plot shows a mild upward trend, suggesting some heteroscedasticity. In the perceived competence model, residual diagnostics are notably clean: the Residuals vs Fitted plot is flat, the Q-Q plot shows minor tail deviations for P2, P15, and P11 without systematic pattern, and no observation crosses the Cook's distance 0.5 boundary. The intention to continue model shows P20 and P16 in the lower-left Q-Q tail, consistent with the Shapiro-Wilk violation in the Low Openness / Explicit Tutorial cell; P3 appears in both the Residuals vs Fitted and Scale-Location panels. In the Residuals vs Leverage plot, P3, P16, and P20 cluster near the boundary without crossing it.

With $n = 7-8$ per cell, no single observation can dominate a cell estimate. Cook's distance is used as the influence diagnostic because it captures the combined effect of leverage and residual size

— an observation can have high leverage (unusual predictor values) without being influential if it falls near the fitted surface, and Cook's distance accounts for both simultaneously. Flags here reflect individual data characteristics rather than the structure of the design.

4.8 FALSE DISCOVERY RATE CORRECTION

The Benjamini-Hochberg procedure was applied separately to two families of interaction p-values: the three subjective outcomes and the three behavioural outcomes. BH was chosen over Bonferroni because Bonferroni controls the familywise error rate by assuming all tests are independent and applying a uniform penalty — in an underpowered sample this additional conservatism would further reduce the ability to detect real effects. BH instead controls the expected proportion of false discoveries among significant results, maintaining greater power at the same false discovery threshold. The two families were kept separate because combining them into a single family would penalise the behavioural tests for the subjective outcomes' p-value distribution; the two sets address conceptually distinct questions and their correction should reflect that. For subjective outcomes, raw interaction p-values of .975, .337, and .795 all adjusted to .975. For behavioural outcomes, raw interaction p-values of .496, .733, and .496 adjusted to .733. No interaction effect survived correction in either family. The tutorial condition main effects on total deaths ($p = .044$) and perceived competence ($p = .046$) were not subject to BH correction as they were not the primary confirmatory tests; they are interpreted as exploratory secondary findings.

4.9 SENSITIVITY ANALYSIS: ENJOYMENT EXCLUDING P16

Participant P16 — a Low Openness / Explicit Tutorial participant who quit the session voluntarily and continues to appear in the leverage region of the enjoyment diagnostic plot in the present dataset — was flagged as potentially influential via Cook's distance. In the full sample ($N = 29$), the interaction effect was $F(1, 25) = 0.001$, $p = .975$, $\eta^2 p < .001$. Excluding P16 ($N = 28$), the interaction shifted to $F(1, 24) = 0.378$, $p = .544$, $\eta^2 p = .016$. Both results are non-significant and the shift in $\eta^2 p$ from effectively zero to .016 is negligible, confirming that P16 does not exert undue influence on the enjoyment model. P16 is retained in all primary analyses.

4.10 EXPLORATORY KENDALL RANK CORRELATIONS

As pre-specified in Chapter 3, Kendall's τ was computed between continuous openness score and primary outcomes — preserving the full scale rather than the median-split dichotomy — across the full sample and then separately by tutorial condition.

Across the full sample ($N = 29$), no correlation between openness score and the three primary outcomes exceeded $|\tau| = .220$. The strongest full-sample associations involved perceived difficulty ($\tau = .240$) and frustration ($\tau = .210$), suggesting a modest tendency for higher-Openness participants to perceive the game as harder and more frustrating.

The split by condition shows a directional pattern consistent with the cell means in Table 4. In the Minimal Tutorial condition ($n \approx 14$), $\tau = -.390$ between openness and perceived competence: higher openness was associated with lower competence. In the Explicit Tutorial condition, that association nearly disappears ($\tau = -.044$). Frustration shows the opposite shift: near zero in the Minimal condition ($\tau = -.041$), positive in the Explicit condition ($\tau = .398$) — higher-Openness participants reported more frustration with explicit guidance. That reversal is consistent with a tutorial-moderated personality effect. P-values are not reported at this subgroup size; these are directional observations for the follow-up study.

4.11 OBSERVER EFFECT AND ECOLOGICAL VALIDITY

The researcher was present throughout all sessions in compliance with welfare monitoring requirements. While the non-intervention protocol was followed strictly — limiting researcher communication to technical and welfare matters — the presence of an observer may have introduced social desirability effects or performance anxiety. These effects cannot be eliminated from the current design. Remote or automated data collection would remove this limitation.

4.12 CHAPTER SUMMARY

This chapter reported the execution and results of a study ($N = 29$) examining whether Openness to Experience moderates the effect of tutorial type on player experience in a Souls-like game. The study slightly exceeded its target sample of 24–28, with cells of $n = 7, 7, 7, 8$.

None of the four hypotheses were supported. No interaction effect survived BH correction. The only significant signal is the tutorial condition main effect on perceived competence ($F(1, 25) = 4.433$, $p = .046$, $\eta^2 p = .150$) — but it comes with a caveat: the randomisation produced a sixfold Souls-like experience imbalance between conditions ($M = 10.79$ vs. 1.67 hours), which is a plausible alternative explanation. When deaths are expressed as deaths per minute to control for session length, that effect also disappears ($p = .131$). What remains is one marginal competence signal, a notable paradox in the tutorial evaluation data (better-rated guidance, worse performance), and directional Kendall reversals consistent with a personality-moderated effect. Exploratory correlations and the sensitivity analysis are in Sections 4.10 and 4.9.

5 DISCUSSION OF DATA AND RESULTS

This study asked whether matching tutorial type to player Openness improves the initial experience of a Souls-like game. The prediction was specific: high-Openness players should do better with minimal tutorials, low-Openness players better with explicit ones, with the interaction term carrying the weight of the hypothesis. None of the four pre-registered hypotheses found statistical support. H1 (Openness main effect) and H3 (the interaction) showed no significant effects on any primary outcome. H4 (mediation) was not testable. H2 produced a significant tutorial main effect on perceived competence, but in the opposite direction predicted, with minimal-tutorial participants reporting higher competence than explicit-tutorial participants.

The interesting question is not whether this result is large enough to act on yet. It probably isn't, given the sample size and the experience confound. The interesting question is what mechanism could produce it, and whether that mechanism is worth a properly powered replication. The bulk of this chapter is concerned with that question.

5.1 THE CENTRAL FINDING: WHY LESS GUIDANCE PRODUCED MORE COMPETENCE

Participants who received minimal tutorial guidance reported feeling more competent than those who received explicit step-by-step instruction. H2 predicted the opposite. The straightforward reading is that explicit instruction directly produced lower competence ratings, but the data also contain a sixfold imbalance in prior Souls-like experience between conditions ($M = 10.79$ vs. 1.67 hours), a randomisation accident with real consequences for interpretation. Both readings need to be taken seriously, and the most informative version of the discussion engages with both.

This is also a result that sits in tension with one of the strongest prior findings in tutorial research. Andersen et al. (2012), working across 45,000 players in three games of varying complexity, found that tutorials improved retention by 29% in complex games. The present finding, in a much smaller sample and with a different outcome, points the other way. The two studies can be reconciled (different definitions of tutorial, different outcomes, different games), but the direction of the disagreement matters: it suggests the relationship between guidance and player outcomes is not as monotonic as the broader literature implies.

5.1.1 The Confound and What It Cannot Explain

The experience imbalance is real. Players with ten hours of Souls-like exposure already know how the genre teaches: cryptic signals, environmental learning, repetition without explicit feedback. It would not be surprising if they felt more competent in a minimal tutorial as a result. Some portion of the competence effect, and most likely a substantial portion of the raw deaths effect, is attributable to this imbalance rather than to the tutorial manipulation itself. The fact

that the deaths effect disappears once expressed as deaths per minute ($p = .131$) confirms this: raw deaths were partly a function of session length, which was itself related to engagement, which was itself related to genre familiarity.

But the confound is incomplete as an explanation. If experience were the whole story, the effect should track gaming experience broadly, not just Souls-like exposure. More importantly, it should be visible in enjoyment and intention to continue, which it isn't. Both measures are flat across all four cells. Genre-experienced participants did not report enjoying the game more, and did not express stronger intention to continue. The competence effect emerges as a specifically self-evaluative judgement that does not propagate to the affective or behavioural-intention measures it would be expected to influence if it were purely a downstream consequence of experience. Something about the explicit-tutorial condition specifically suppressed how competent participants felt without making them enjoy the game less or want to play it less. That asymmetry is the part the confound does not easily explain.

5.1.2 Three Mechanisms Worth Testing

Three mechanisms could plausibly produce the asymmetry observed. None of them are confirmed by this study; the sample size does not support causal claims. They are the candidate hypotheses a properly powered replication would need to discriminate between.

The first is autonomy-undermining. Self-Determination Theory (Ryan & Deci, 2000) predicts that perceived competence is not just a function of actual performance, it is a function of how that performance was achieved. Solving a problem oneself produces a different competence signal than executing a solution that was provided. If the explicit tutorial pre-loaded participants with the answers, then every later success was partially attributable to the guide rather than to the player. The same boss defeat, the same enemy parried, the same mechanic understood, but the credit lands differently depending on whether the player worked it out. This mechanism would predict exactly what the data show: a competence drop without a corresponding enjoyment drop, because the game is still enjoyable to play, but it stops feeling like a personal accomplishment.

The second is expectation-anchoring. The explicit tutorial set a baseline of what participants should be able to do once they had read it. Failure against that baseline carries more weight than failure without it. A minimal-tutorial player who dies to the first enemy has no internal model telling them how well they should be doing; they recalibrate as they go. An explicit-tutorial player has just spent five minutes reading about stagger mechanics, perfect guards, and Ergo recovery, and then dies to the first enemy anyway. The gap between the prepared player they expected to be and the player they actually were is a measurable cost. This mechanism would predict the tutorial evaluation paradox directly: participants rate the guidance as more appropriate ($M = 4.20$ vs. 3.71) because it covered the right material, but they feel less

competent because the material did not translate into performance. Both judgements can be accurate simultaneously.

The third is cognitive overhead at the worst moment. Reading a five-minute guide immediately before play loads short-term memory with terminology, control mappings, and conceptual frameworks that have not yet been grounded in any experience. The minimal-tutorial participant arrives at the first combat encounter with attention available for the encounter. The explicit-tutorial participant arrives at the same encounter trying to remember which input was R2, what the stagger bar looked like, and how perfect guards differed from regular blocks, none of which has been practiced. The cognitive load of the guide may itself have degraded early performance, with the competence drop reflecting the felt experience of trying to recall guidance under pressure rather than learning through doing. This mechanism is testable: delaying the guide, breaking it into in-context tooltips, or providing it after a brief unguided session would isolate it from the other two.

The three are not mutually exclusive. The first two describe what the explicit tutorial does to participants psychologically; cognitive overhead describes what it does to them mechanically. The first two would persist even if the guide were perfectly retained; the third would partly resolve with better delivery. For developers, this is the actionable distinction: front-loaded text dumps before play are the worst possible format, because they maximise cognitive overhead, which is the only one of the three mechanisms that delivery design can directly reduce. Just-in-time tooltips, in-context prompts at first encounter, or short interactive practice segments would all target that mechanism without changing what the tutorial covers, while leaving the two psychological effects in place. A study capable of separating all three would need to vary tutorial intensity, timing, and framing, a three-way design that is the natural next step from this one.

5.1.3 The Tutorial Evaluation Paradox

Independent of mechanism, the paradox in the tutorial-evaluation data deserves direct attention. Explicit-condition participants gave the guide higher ratings on its appropriateness, closer to the "just right" midpoint of the scale, than minimal-condition participants gave to their lack of guidance. The same participants who rated the guidance as well-calibrated reported feeling less competent and died more often than those who did not receive it. The metacognitive signal is moving in the opposite direction to the performance signal.

This is consistent with the autonomy-undermining and expectation-anchoring readings, and it carries a practical implication independent of which mechanism is correct: players are not good judges of what helps them. If a developer asks players to evaluate a tutorial, they will report the tutorial that covered more material as the better tutorial, even when that tutorial measurably degraded their performance and self-rated competence. The standard playtesting question ("Was the tutorial helpful?") may be measuring something other than tutorial helpfulness. The

participants are not lying or confused; they are correctly evaluating what the guide *covered*, which is not the same thing as evaluating how the guide *worked*. Distinguishing these two questions is something the field could usefully take more seriously.

5.2 WHAT THE INTERACTION NULL AND THE KENDALL REVERSAL TELL US TOGETHER

H3, the matching interaction, was the confirmatory test, and it found nothing. The interaction p-values across the three subjective outcomes were .975, .337, and .795, all well above the threshold even before BH correction. On the headline test, the personality-tutorial fit hypothesis received no support. This is consistent with the broader pattern in personality-gaming research: Akbari et al.'s (2021) meta-analysis of Big Five effects in online gaming reports generally small associations, and the present null sits comfortably inside that picture rather than contradicting it.

But two features of the data complicate the cleanest reading of that null. First, the exploratory Kendall correlations split by condition show a clear directional reversal: $\tau = -.390$ between Openness and perceived competence in the minimal condition, $\tau = -.044$ in the explicit condition. Higher Openness predicted lower competence specifically when guidance was withheld, a pattern the interaction should have caught if it had the power to. Frustration shows the same shape: $\tau = -.041$ minimal, $\tau = .398$ explicit. Higher-Openness participants reported more frustration with explicit guidance, not less. Both correlations move in the directions a personality-moderated effect would produce. Neither is large enough to survive a formal subgroup test at $n \approx 14$ per cell, but both are too directionally consistent to ignore.

Second, the descriptive pattern in the cell means is also asymmetric in the right direction. Low-Openness participants dropped 1.85 competence points when moved from minimal to explicit, while High-Openness participants dropped only 0.68 points. That is the direction predicted by the matching hypothesis (the lower-Openness group suffered more from the wrong tutorial), even though the interaction term itself did not reach significance.

The honest reading is that the interaction null and the directional signal are not contradictory. They describe the same data at different levels of resolution. The interaction test is a single statistic asking whether the effect of tutorial type differs cleanly between two roughly balanced groups of 14 and 15 participants. With cells of seven and eight, it can only detect interactions large enough to be visible without statistical aid. The Kendall split preserves the continuous Openness measure and looks at the correlation within each condition rather than between condition means. It is a weaker test in terms of inference, but a more sensitive one in terms of pattern detection. McClelland and Judd (1993) make this point directly: interaction tests in unbalanced or small samples lose power well below what main-effect calculations suggest, and

a small interaction is exactly the kind of effect that would produce significant Kendall reversals without producing a significant ANOVA interaction.

What this study cannot say is whether the personality-moderated effect exists at the population level. What it can say is that the data are not inconsistent with one: the cell means lean the right way, the correlations lean the right way, and the null result is fully accounted for by the sample size. That is the version of the finding that justifies a powered replication.

The H1 main effect of Openness produced no significant or directional result on any primary outcome. Openness on its own did not predict enjoyment, competence, or intention to continue. This is consistent with the conceptual picture: Openness should not produce a main effect because its predicted effect is contingent on the tutorial condition. A high-Openness player in the wrong tutorial should look indistinguishable from a low-Openness player in the right one. Main effects of Openness on these outcomes were not the prediction; their absence is uninformative either way. The Komarraju et al. (2011) finding from the education literature, that high-Openness students learn more effectively under discovery-based instruction, supports treating Openness as a moderator rather than a main predictor: even in a domain with much larger samples and longer outcome windows, the effect operates through interaction with instructional style rather than independently of it.

5.3 ENJOYMENT AS A STABLE FLOOR

Enjoyment ratings spanned only 0.57 scale points across the four cells, with no systematic pattern by Openness or condition. Whatever else the manipulation did, it did not change how much participants enjoyed playing the game. This is more useful than a null usually is. It rules out the simplest alternative explanation for the competence reversal: that explicit-condition participants felt less competent because they generally had a worse time. They didn't. They enjoyed the game equally and intended to continue playing it equally. The competence signal is not riding on top of a general affective disturbance; it is doing something more specific. It also suggests that *Lies of P* is robust to onboarding variation in a way the broader Souls-like genre may share. The core loop (death, learning, repeated attempts, eventual breakthrough) is engaging enough that tutorial design at the margin does not dent it. This may be partly self-selection: participants who agreed to a 90-minute session with a Souls-like game are not the general population. But within that population, the affective experience appears stable in a way that the self-evaluative experience does not. Players can have a good time and still feel that they were not very good at it.

5.4 METHODOLOGICAL CONSIDERATIONS

Three methodological points shape how much weight the findings carry. The experience imbalance is the most consequential. Random assignment within Openness strata produced

groups that differed by approximately nine hours of prior Souls-like exposure. With cells of seven and eight, a single experienced participant ending up in one condition rather than the other can swing the group mean noticeably, and that is what happened. Block randomisation on prior genre experience (assigning the most-experienced participant to one condition, the next to the other, and so on) would have prevented this and is the obvious correction for any replication. Reporting the analyses with experience as a covariate is also feasible, but at this sample size adding a covariate to an already underpowered model is not a clean solution; it is a partial correction that introduces its own assumptions.

The deaths-per-minute clarification deserves restatement. The raw total-deaths effect was significant and pointed the same direction as competence, but it dissolved once session length was controlled for. This is the right way to read a count outcome over a variable-duration session (participants who played longer had more opportunity to die), and it removes deaths from the inferential picture. What remains is competence as the only outcome with both a significant tutorial effect and an interpretable mechanism. That is one outcome out of three primary measures, and it should be reported as such rather than as part of a converging set.

The observer effect is unmeasurable from inside the design. Participants knew they were being watched, knew the session was recorded, and may have modulated their behaviour and self-report accordingly. The non-intervention protocol minimises the researcher's direct influence, but presence itself is the confound. Remote, asynchronous data collection (participants playing in their own homes on their own time, with telemetry and post-session questionnaires) would remove this entirely and would be the appropriate vehicle for a powered study.

5.5 WHAT THIS MEANS FOR DESIGNERS

The findings do not justify changes to commercial tutorial design. The sample is too small, the confound too consequential, and the interaction effect too speculative for that. But they do produce two ideas worth taking into the design conversation.

The first is that explicit pre-play guidance may be a worse default than it looks. In playtesting, players will reliably rate more thorough tutorials as better tutorials, but the same players may report feeling less competent and may actually perform worse than players who received less guidance. The standard methodology of asking players what they thought of the tutorial is measuring satisfaction with what was covered, not the effect of that coverage on competence or performance. Developers who care about the gap between satisfaction and effect should consider measuring competence separately, ideally through self-report and behavioural measures collected without the player being asked to evaluate the guidance directly.

In practice, this means tracking objective markers that do not depend on the player's evaluation: time to first checkpoint, retries before progression, damage taken in the first combat encounter, and how much of the starting area was explored. These move independently of how

the player rates the tutorial, and they are closer to what actually predicts whether the player will keep playing.

The second is that personality-adaptive design remains worth investigating, but probably not at the granularity of "high-Openness players should get a minimal tutorial." The Kendall correlations suggest the effect is continuous rather than categorical, and the matching may operate at a finer level than tutorial intensity (perhaps in how mechanics are framed, when guidance appears relative to first encounter, or how the game communicates that the player has made progress). A future-state implementation might detect personality not through a questionnaire (which players will not tolerate before play) but through early behavioural signals: who reads versus skips item descriptions, who explores versus pushes forward, who experiments with mechanics versus follows the prompted path. This is the direction Denden et al. (2018) take, and it is a meaningfully different research programme than the present study, which tests the ceiling case of direct measurement.

There is some commercial precedent for asking the questionnaire directly. *Pokémon Mystery Dungeon: Red Rescue Team / Blue Rescue Team* and *Dragon Quest III* both put short personality quizzes at the start of the game, and players engage with them as a feature rather than a chore. They show that the format works at character-creation scale. They also expose the honesty problem: when players can see that their answers shape the game, they answer to optimise rather than to reflect. For tutorial adaptation this is amplified. If players can tell that the tutorial length depends on their answers, they will simply select whichever option produces the shorter tutorial, regardless of what their personality actually is. Behavioural inference avoids this. Watching how someone plays the first five minutes cannot be gamed in the same way.

5.6 THE BIGGER QUESTION

The most useful way to read this thesis is not as a study that failed to confirm its hypothesis. It is as a study where the prevailing intuition about how to onboard players produced the wrong prediction. The intuition is widely held: more guidance produces a more confident, more capable player. It underwrites most commercial tutorial design. The data did not just fail to support it. They pointed the other way. And the same players who received more guidance also rated that guidance as more appropriate. They believed it helped. It did not. That gap, between what players think helped them and what actually helped them, is the part worth taking outside the Souls-like context.

If a theoretically reasonable prediction about instructional design can be this directionally wrong, the obvious next question is what else might be. Some candidates: that longer onboarding produces more competent players; that explaining mechanics before they appear reduces frustration; that step-by-step guidance is universally more effective than discovery

learning for novices; that "comprehensive" tutorials are better than minimal ones. Every one of these is intuitive. None of them have been tested with the rigour the assumptions deserve.

The stakes climb sharply outside games. The tutorial-evaluation paradox observed here is not specific to entertainment. It generalises to any training context where the learner's self-assessment is part of the outcome. Consider any high-stakes domain where someone receives a tutorial before performing a task that matters: a junior clinician learning a procedure, a new driver learning road rules, an operator learning a safety system. If the people being trained systematically rate their guidance as more appropriate than it actually was, and self-rate their competence as higher than their performance warrants, the training produces a confident person who is less capable than they believe themselves to be. That gap is dangerous in exactly the contexts where competence matters most. Consider a junior pilot trainee who, after a thorough pre-flight briefing, reports both that the briefing was well-calibrated and that they feel ready to fly. They then perform measurably worse in the simulator than a peer who received a shorter briefing. If both trainees subsequently believe themselves equally capable, the more-briefed one is the more dangerous of the two. That is the shape of the risk: confidence calibrated to coverage rather than capability.

The present study cannot prove this generalises. It is a ninety-minute session, one game, twenty-nine participants. But it can sharpen the question. The literature on training design across domains has historically treated subjective satisfaction with the training and objective effectiveness of the training as if they were the same measurement. The data here suggest they are not, at least in one specific high-difficulty context. The broader research question is whether they diverge elsewhere too, and where the divergence is largest.

That is what the thesis lands on, even if it cannot answer it. The personality-matching hypothesis was a specific test of a specific idea. The result is interesting less for what it says about Openness and tutorials, and more for what it suggests about how much of our intuition about onboarding might be wrong in ways we have not yet checked.

5.7 CHAPTER SUMMARY

The study produced one statistically significant finding in the wrong direction (lower competence with explicit tutorials), one suggestive directional pattern (Kendall reversals consistent with a personality-moderated effect), one informative null (flat enjoyment ruling out general affect as a confound), and one methodologically consequential imbalance (sixfold prior Souls-like experience between conditions). Together they do not constitute support for the personality-tutorial fit hypothesis as originally specified. They do constitute a coherent enough pattern to justify a powered replication with block randomisation on prior genre experience and a design capable of separating autonomy-undermining, expectation-anchoring, and cognitive-overhead explanations of the competence reversal. More broadly, the finding speaks

to how much of our intuition about onboarding may be wrong in ways that have not been checked, in games and outside them. The next chapter draws the practical implications together and outlines what such a replication would look like.

6 CONCLUSION AND FUTURE DIRECTIONS

6.1 CONCLUSION

This thesis tested whether matching tutorial intensity to player Openness improves the initial experience of a Souls-like game. The personality-by-tutorial interaction predicted by H3 did not emerge, and the only significant tutorial main effect ran against H2: minimal-tutorial participants reported higher perceived competence than explicit-tutorial participants. A sixfold imbalance in prior Souls-like experience between conditions, produced by random assignment at $N = 29$, partially confounds this result and means it cannot be taken at face value. The directional Kendall reversals between conditions are consistent with a personality-moderated effect the study lacked power to confirm. The findings are best read as a feasibility study: the measurements work, the manipulation produces detectable effects, and the most interesting effect runs in a direction the original framework did not anticipate. The lessons for future research are concrete: pre-stratify on prior genre experience, deliver tutorials in-game rather than externally, and measure performance alongside satisfaction rather than collapsing the two.

6.2 CONTRIBUTIONS

The contributions of this work are not contingent on supporting the original hypotheses. Three are worth stating explicitly. First, this is the first empirical test of a personality-by-tutorial interaction in the Souls-like genre. The hypothesis was theoretically well-motivated by Person-Environment Fit theory (Edwards, 2008), Self-Determination Theory (Ryan & Deci, 2000), and prior work on Openness and learning style (Komarraju et al., 2011), but had not been tested directly in any high-difficulty game context before now. The null interaction reported here defines a floor for the magnitude of any real effect: it is smaller than this design could detect at $N = 29$. Second, the H2 reversal, taken together with the tutorial-evaluation paradox, suggests that the standard playtesting question of whether a tutorial is "helpful" measures something other than tutorial effectiveness. That is a practical observation game developers can act on independent of any replication. In concrete terms: during playtesting, supplement subjective tutorial-quality ratings with objective performance measures from the first hour of play, and treat any divergence between the two as a signal that the tutorial may be miscalibrated. Third, the methodological reflection in this thesis provides a working recipe for the next study (stratified randomisation on gaming experience, rate-adjusted behavioural outcomes, an in-game rather than external tutorial manipulation, and a three-way design capable of discriminating between the candidate mechanisms) concrete enough to be lifted directly into a replication protocol.

6.3 LIMITATIONS

The limitations concentrate around four standard dimensions: internal validity, external validity, reliability, and replicability.

Internal validity is the principal concern. The sixfold imbalance in prior Souls-like experience between tutorial conditions cannot be cleanly separated from the tutorial manipulation in the present design, and it provides a plausible alternative explanation for the competence and raw-deaths findings. Block randomisation on prior genre experience would resolve this and is the obvious methodological correction. The non-blind researcher observation is the second internal-validity issue: participants knew they were being watched, and the explicit condition combined supplementary guidance with the visible expectation that comes with it. Variable guide compliance is the third: tutorial exposure was not uniform within the explicit condition, which weakens the comparison in both directions.

External validity is limited by sample and scope. With cells of seven and eight, only large effects could be detected; the small-to-medium interactions the matching hypothesis would plausibly produce would be invisible to the inferential tests used. Results are specific to *Lies of P* and to Western participants. The game was chosen for its representativeness within the Souls-like genre, but it is one title among many, and the early-game dynamics it presents are not perfectly interchangeable with those of *Dark Souls* or *Elden Ring*. Cultural background was deliberately controlled by limiting recruitment to Western participants; whether the pattern holds in East Asian markets is a separate question. The 90-minute session captures initial experience only. The hypothesis concerns early abandonment, which is the right scope, but the study cannot speak to longer-term retention. Intention to continue is the available proxy, and it was flat across conditions.

Reliability is bounded by the personality measure. The BFI-10 captures Openness with two items and acceptable but not high reliability ($\alpha \approx .73$). A noisier signal at the predictor end will attenuate any interaction at the outcome end, even when the underlying interaction is real. The present study cannot tell whether the null interaction reflects no effect or a real effect blurred by measurement noise on top of all the other small-sample problems. A larger replication should use a longer scale such as the BFI-44 or NEO-FFI.

Replicability is supported by the pre-registered analysis plan and the documented deviations from it in Section 4.2, but constrained by the same small-sample issues. The randomisation procedure was followed correctly and produced a sixfold covariate imbalance anyway. Anyone running the same procedure with $N \approx 29$ should expect to confront the same risk. Stratified randomisation, rather than simple random assignment, is the practical fix.

6.4 FUTURE DIRECTIONS

The most direct next step is a powered replication. A sample of approximately 120–160 participants (calculated from the $\eta^2p = .150$ competence effect and the directional Kendall pattern, with reasonable allowances for the effect shrinking once the experience imbalance is removed) would provide adequate power to detect a medium-sized interaction. The replication should use block randomisation on prior Souls-like experience, ideally with stratification on both Openness and experience to avoid the imbalance encountered here.

The replication should also be designed to discriminate between the three candidate mechanisms for the competence reversal. Manipulating tutorial timing (before play vs. during play vs. after a brief unguided session), tutorial format (text vs. annotated screenshots vs. in-context tooltips), and explicit vs. implicit framing of mechanics would isolate cognitive overhead from autonomy-undermining and expectation-anchoring. A three-arm design rather than a two-arm one would do more work per participant than scaling up the original 2×2 .

Remote data collection is the next methodological priority. Participants playing in their own environment, recording telemetry through the game's existing data infrastructure, and completing post-session questionnaires asynchronously would remove the observer effect and would expand the sample beyond what in-person sessions allow. Souls-like games are particularly suited to this because their telemetry (deaths, encounters, time-to-progress) is already tracked by the engine.

Conscientiousness is the second personality dimension the literature predicts should matter for Souls-like persistence (Roberts et al., 2009), and the natural next study would test Openness and Conscientiousness together. A two-trait design needs a larger sample than this thesis used (interactions between two continuous traits require approximately three times the power of single-trait designs), but it is the version of the question Person-Environment Fit theory (Edwards, 2008) actually implies.

Beyond the immediate replication, the longer-term direction is detection without questionnaires. Players will not tolerate a Big Five inventory before play in any commercial context, so personality-adaptive onboarding will only become practical if personality (or a behavioural proxy for it) can be inferred from early actions: time spent reading item descriptions, ratio of exploration to direct progression, frequency of mechanic experimentation. Item-description reading time likely tracks Conscientiousness more than Openness; exploration ratio is plausibly the cleaner Openness signal; mechanic experimentation may sit between the two. Whether early behavioural signals predict tutorial preferences as well as a personality inventory would itself answer a useful question. If they do, the implementation path opens up considerably; if they don't, personality-adaptive onboarding remains a research curiosity rather than a design tool.

6.5 CLOSING REMARKS

The Cody and May scenario from the opening of this thesis described two players with identical games and different experiences. This study did not produce a clean account of why that difference exists, but it produced a more useful one than a confirmation of the original hypothesis would have. The data point toward an effect that is real, smaller than expected, more specific than expected, and running in a direction that complicates the simple story. Players given more guidance felt less competent, not because the guidance was bad, but possibly because of what guidance does to the experience of figuring things out. That is a more interesting result than the matching hypothesis would have been if it had been confirmed cleanly. It is also a result that opens more questions than it closes, which is the appropriate outcome for a pilot study and the appropriate starting point for the work that comes next.

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